From Static Screening to Autonomous Discovery: A Paradigm Shift in Data-Driven Electrocatalysis

The discovery of next-generation electrocatalysts is undergoing a fundamental transition from serendipitous trial-and-error to rational, data-driven design, necessitated by the combinatorial explosion of chemical space. This review synthesizes recent advancements where machine learning (ML) and artificial intelligence (AI) leverage massive infrastructure like [77] and [21] to navigate vast material landscapes. We analyze the evolution from static descriptor-based screening to dynamic inverse design using generative frameworks such as [1] and agentic systems like [94], which actively break conventional scaling relations. Crucially, we address the tension between predictive accuracy and physical interpretability, highlighting hybrid approaches like [195] that embed domain knowledge into neural architectures. Furthermore, the review examines the critical "simulation-to-experiment" gap, discussing how active learning and self-driving laboratories, exemplified by [39], close the loop between computation and synthesis. By integrating high-fidelity datasets, generative modeling, and autonomous workflows, this field is moving toward fully autonomous discovery ecosystems capable of accelerating sustainable energy technologies.

Introduction: The Paradigm Shift to Data-Driven Electrocatalysis

The integration of machine learning (ML) and artificial intelligence (AI) into electrocatalyst discovery marks a fundamental transition from serendipitous, trial-and-error experimentation to rational, data-driven design. This paradigm shift is necessitated by the combinatorial explosion of chemical space, which renders traditional exhaustive screening via density functional theory (DFT) computationally prohibitive. For instance, exploring the vast topological and compositional configurations of materials such as MXenes or high-entropy alloys involves design spaces that can exceed millions of candidates, a scale that is intractable for routine theoretical approaches or sequential experimental testing [50; 4]. By leveraging ML, researchers can now navigate these immense landscapes efficiently, approximating quantum mechanical accuracy at a fraction of the computational cost and enabling the rapid identification of candidates for critical reactions such as the hydrogen evolution reaction (HER), oxygen evolution reaction (OER), and CO2 reduction.

Central to this transformation is the development of robust data infrastructure that serves as the foundation for training generalizable models. Historically, catalyst discovery was hindered by fragmented, small-scale datasets that limited model transferability across different material classes. The advent of large-scale, standardized repositories has begun to dismantle these barriers. The Open Catalyst Project, through the release of the OC20 and OC22 datasets, has provided foundational benchmarks comprising millions of DFT relaxations across diverse metals and oxides, facilitating the training of graph neural networks that generalize across elemental compositions and adsorbate configurations [77; 21]. Complementing these efforts, databases like BEAST DB offer grand-canonical electrochemical data computed in implicit solvent, allowing models to learn higher-fidelity properties under realistic operating conditions rather than relying solely on static vacuum calculations [11]. These resources collectively enable a move away from specialized, narrow models toward universal potentials capable of accelerating discovery across a variety of applications without the need for specialized retraining for every new chemistry [213].

Beyond mere acceleration of screening, AI is fundamentally inverting the design process through generative models and autonomous agents. Traditional workflows typically screen existing libraries for promising candidates; in contrast, generative frameworks such as Catalyst GFlowNet and CatGPT leverage probabilistic and language model architectures to propose entirely novel crystal structures and alloy compositions tailored to specific target properties [1; 34]. This capability is further amplified by the emergence of agentic AI systems, where large language models (LLMs) act as autonomous researchers. Systems like Catalyst-Agent and MAESTRO utilize multi-agent architectures to reason through chemical space, hypothesize design rules, and execute screening workflows with minimal human intervention, effectively breaking conventional scaling relations that have long constrained catalyst performance [94; 7; 56]. These developments suggest a future where AI does not merely assist but actively drives the scientific method, integrating textual knowledge from literature with quantum-chemical feedback to navigate complex design spaces.

However, the realization of this data-driven future relies on closing the loop between computation and experiment. Black-box models, while accurate, often fail to provide the physical insights necessary for rational design, prompting a surge in explainable AI (XAI) and theory-infused neural networks that embed physical laws, such as d-band theory, directly into the learning architecture [195; 90]. Furthermore, the disconnect between static computational models and dynamic experimental realities is being addressed through active learning loops and automated workflows. Frameworks like AutoMat and multi-stage ML-driven approaches integrate domain adaptation and active learning to iteratively refine models with experimental feedback, minimizing the number of required experiments while maximizing information gain [39; 3]. As these tools mature, they promise to democratize access to advanced catalyst design, creating a virtuous cycle of discovery that accelerates the deployment of sustainable energy technologies required to address global climate challenges [78; 40].

The Imperative for Sustainable Energy Conversion

The global urgency to decarbonize energy systems necessitates a fundamental paradigm shift away from noble metal-dependent catalysis toward earth-abundant, high-performance alternatives capable of operating under industrially relevant conditions, such as high current densities and harsh electrolytes. Traditional approaches relying on platinum group metals are increasingly viewed as unsustainable due to their scarcity, high cost, and vulnerability to degradation mechanisms such as leaching and surface reconstruction. Consequently, the search for next-generation electrocatalysts has expanded to include single-atom catalysts (SACs), high-entropy alloys (HEAs), topological quantum materials, and defect-engineered nanostructures. Achieving carbon neutrality requires not only catalysts with high intrinsic activity but also those possessing exceptional stability and selectivity, demanding a multiscale design strategy that integrates morphology, surface chemistry, and degradation mechanisms [72]. However, the sheer complexity of these new material classes renders traditional trial-and-error or purely physics-based computational approaches insufficient, creating a critical need for data-driven methodologies to navigate the vast chemical and structural spaces.

A primary challenge in the rational design of these advanced materials is the unreliability of conventional descriptors when applied at scale. While adsorption energies and thermodynamic overpotentials have long served as primary screening metrics, recent large-scale analyses suggest they are necessary but insufficient for identifying superior catalysts. Uncertainties in density functional theory (DFT) calculations, often ranging from 0.3 to 0.5 eV, can lead to the misclassification of a broad fraction of materials, limiting the utility of overpotential as a standalone sorting criterion [31]. This limitation is particularly acute for complex systems where static models fail to capture the dynamic nature of active sites under operating conditions. For instance, in bimetallic systems like Pd-Au, degradation is driven by complex atomistic evolutions, including the leaching of specific components and surface segregation, which static adsorption energy calculations fail to capture fully [60]. Advanced characterization techniques and machine-learned force fields have revealed that active sites are dynamic, evolving under reaction conditions in ways that static models cannot predict, necessitating a shift toward simulations that account for thermal treatment and reactive atmospheres [115].

To address these complexities, researchers are increasingly turning to compositionally complex solid solutions and high-entropy alloys (HEAs), which offer a vast compositional space for tuning electronic structures and breaking linear scaling relations. The combinatorial explosion of possible compositions in HEAs makes exhaustive DFT screening prohibitive, positioning machine learning as an essential tool for optimization. Machine learning models trained on high-throughput data have successfully identified specific HEA compositions, such as Co-Mo-Fe-Ni-Cu, that rival the activity of ruthenium for ammonia decomposition while utilizing earth-abundant elements [120]. Furthermore, the strategic exploitation of surface segregation in bimetallic nanoparticles allows for the enrichment of noble metals on the surface, significantly reducing usage while maintaining activity and stability, a phenomenon that can be predicted and optimized using thermodynamic descriptors derived from molecular dynamics simulations [249]. Similarly, controlling metastability through annealing in high-entropy nanoalloys has been shown to enhance oxygen evolution reaction (OER) activity by forming heterostructured nanoparticles with reinforced carbon shells, demonstrating that synthesis conditions are as critical as composition [235].

Beyond alloying, the emergence of topological quantum catalysts (TQCs) offers a novel mechanism for enhancing catalytic performance through robust surface states and high carrier mobility, representing a departure from traditional d-band theory. Two-dimensional materials with nontrivial band topology, such as nodal-line semimetals and topological insulators, exhibit metallic edge states that facilitate charge transfer and yield Gibbs free energies for hydrogen adsorption comparable to platinum [Topological quantum catalyst: the case of two-dimensional traversing nodal line states associated with high catalytic performance for hydrogen evolution reaction]. Theoretical studies on materials like LiMgAs and A3B compounds demonstrate that topological polarization and Fermi arcs can significantly lower reaction barriers, providing a pathway to ultra-high performance without noble metals [246] [102]. Additionally, spatial structure engineering, such as the development of mosaic catalysts, offers a geometric degree of freedom to enhance mass transfer and local electric fields, further boosting performance beyond what chemical composition alone can achieve [22].

Single-atom catalysts supported on diverse substrates, including graphene, MXenes, and covalent organic frameworks, represent another frontier in maximizing atom utilization, yet their stability remains a paramount concern. The coordination environment of the metal center plays a decisive role in tuning activity; for example, dual-atom catalysts with engineered C/N coordination have shown bifunctional capabilities for overall water splitting [Coordination Engineering of Dual-Atom Catalysts for Overall Water Splitting]. Moreover, the introduction of ligands or dopants can restructure the local microenvironment, as seen in halogen-liganded Pd on MoS_2, which drastically improves hydrogen evolution reaction (HER) performance [152]. However, the stability of these isolated sites requires rigorous electrochemical stability analysis alongside activity screening to filter out candidates prone to poisoning or dissolution under operating conditions, highlighting the need for descriptors that integrate thermodynamic stability with kinetic activity [63]. The integration of data-driven approaches, including generative models and hierarchical screening, is accelerating the discovery of these advanced materials by exploring vast chemical spaces to identify stable electrides and other exotic compounds with tailored electronic properties [64]. Ultimately, the transition to sustainable energy conversion relies on a holistic design paradigm that couples atomic-scale precision with multiscale engineering and robust computational validation to deliver catalysts that are active, stable, and scalable.

Evolution of Computational and Experimental Workflows

The trajectory of computational and experimental workflows in electrocatalysis has undergone a profound transformation, shifting from static, descriptor-based screening paradigms to dynamic, multi-fidelity ecosystems that integrate density functional theory (DFT), machine learning potentials (MLPs), and autonomous robotic experimentation. Historically, the field relied heavily on identifying simple physicochemical descriptors to predict catalytic activity, often limiting the scope to specific material classes where linear scaling relations held true. Early successes in this domain were exemplified by the development of machine learning models capable of predicting perovskite catalytic properties using trivial-to-calculate elemental features, which proved more accurate and significantly faster than models dependent on ab initio-derived features [74]. Similarly, the distillation of accurate descriptors from multi-source experimental data using symbolic regression approaches like SCMT-SISSO allowed researchers to overcome data inconsistencies and identify highly active perovskite oxygen evolution reaction (OER) catalysts from vast chemical spaces [61]. These foundational efforts demonstrated that interpretable descriptors could be engineered to guide screening, yet they often struggled with generalizability across diverse chemical environments or when faced with the high intricacy of atomistic processes governing stability [42].

To address the limitations of static descriptors and the prohibitive computational cost of high-fidelity DFT calculations for large-scale screening, the community has increasingly turned toward large-scale datasets and machine learning potentials. The curation of the Open Catalyst 2020 (OC20) dataset marked a pivotal moment, providing over a million DFT relaxations to train universal machine learning potentials capable of generalizing across elemental compositions and adsorbate configurations [77]. This expansion continued with the Open Catalyst 2022 (OC22) dataset, which specifically targeted oxide electrocatalysts, addressing a critical gap in training data for OER applications and enabling models to capture intricate long-range electrostatic interactions [21]. Recent advancements have shown that learned force fields are now sufficiently robust to replace DFT in ground state catalyst discovery, often finding structures with lower energies than traditional DFT relaxations while operating orders of magnitude faster [281]. Furthermore, specialized databases such as BEAST DB have emerged, offering grand-canonical electrochemical data computed in implicit solvent to facilitate the training of models that can predict higher-fidelity properties under realistic operating conditions [11].

The integration of these data resources has catalyzed the development of autonomous AI agents that can hypothesize, synthesize, and test materials with minimal human intervention. Frameworks like AutoMat illustrate the power of automated workflows that manage simulations across scales, from first principles to continuum modeling, while seamlessly integrating machine learning surrogates and robotic experiments "in-the-loop" to optimize device performance [39]. Building on this, agentic systems powered by large language models (LLMs) have begun to operationalize catalyst screening. For instance, Catalyst-Agent utilizes an LLM to explore material databases, perform structural modifications, and calculate adsorption energies via graph neural networks in a closed-loop manner, achieving high success rates in identifying viable candidates for reactions such as oxygen reduction and CO2 reduction [94]. Similarly, the MAESTRO framework employs multi-agent LLMs to iteratively reason and propose modifications, successfully discovering single-atom catalysts that break conventional scaling relations through in-context learning [7]. These agents represent a shift from passive prediction tools to active scientific collaborators capable of navigating complex design spaces.

Despite these advances, challenges remain regarding the balance between model accuracy, interpretability, and generalizability. While deep learning models like graph neural networks offer superior predictive performance, they often function as black boxes, obscuring the physical insights necessary for rational design. To bridge this gap, hybrid approaches such as theory-infused neural networks (TinNet) integrate deep learning with established physical theories like d-band theory, ensuring predictions are both accurate and physically interpretable [195]. Additionally, active learning strategies have become essential for navigating vast compositional spaces efficiently, particularly for complex systems like high-entropy alloys where traditional trial-and-error is infeasible. Methods utilizing uncertainty quantification, such as Convex hull-aware Active Learning (CAL), prioritize experiments that minimize uncertainty in thermodynamic stability, dramatically reducing the number of required observations [286]. The evolution toward self-driving laboratories further cements this trend, where AI-driven workflows coordinate high-throughput experimentation and continuous flow chemistry to create a virtuous circle of data generation and model refinement [40].

As the field progresses, the convergence of generalizable large-scale models, interpretable physics-informed architectures, and autonomous experimental platforms promises to accelerate the discovery of next-generation electrocatalysts for sustainable energy applications.

Data Infrastructure, Descriptor Engineering, and Representation Learning

The efficacy of machine learning (ML) in catalysis is intrinsically linked to the symbiotic evolution of data infrastructure and representation learning strategies. Early computational approaches relied heavily on hand-crafted descriptors, such as the d-band center, which, while physically intuitive, often failed to capture the complex, non-linear structure-property relationships inherent in diverse catalytic systems. These simplified descriptors struggled to generalize across the vast chemical space required for universal catalyst discovery. Consequently, the field has pivoted toward the curation of standardized, large-scale databases that enable the training of models capable of interpolating and extrapolating across elemental compositions and adsorbate configurations. The release of the Open Catalyst 2020 (OC20) Dataset marked a paradigm shift, providing over a million density functional theory (DFT) relaxations to facilitate the development of models that generalize across materials and adsorbates, addressing the historical limitation of small, chemistry-specific datasets [77]. This infrastructure supports the construction of universal machine learning potentials, though challenges remain in ensuring these models perform robustly across underperforming subsets of chemical space and off-equilibrium data [213]. Further expanding this landscape, the Open Catalyst 2022 (OC22) Dataset specifically targets oxide electrocatalysts, incorporating intricate long-range electrostatic and magnetic interactions essential for Oxygen Evolution Reaction (OER) modeling, thereby complementing the metallic focus of its predecessor [21]. Similarly, specialized repositories like BEAST DB offer grand-canonical electrochemical data computed at consistent parameters, rationalizing trends in catalyst activity and enabling high-fidelity model training under realistic electrochemical conditions [11].

Parallel to database expansion, significant advancements have been made in descriptor engineering and representation learning, moving from human intuition to algorithm-derived features. Modern approaches increasingly utilize Graph Neural Networks (GNNs) to encode local atomic environments, yet the construction of these graphs remains non-trivial due to ambiguous connectivity schemes. Innovations such as Voronoi tessellation have been employed to enrich graph representations by defining edge features based on contact solid angles and node characteristics based on Voronoi volumes, significantly reducing prediction errors on intermetallic datasets compared to standard connectivity definitions [104]. To further enhance generalizability and efficiency, frameworks like PhAST introduce task-specific innovations in graph creation and atom representations, improving energy prediction accuracy while drastically reducing computational costs, thus making large-scale screening more accessible [52]. Beyond standard GNNs, the integration of geometric deep learning with lightweight architectures has demonstrated that robust design patterns, such as symmetric message passing, can rival established models like SchNet and DimeNet++ while utilizing a fraction of the trainable parameters, thereby democratizing access to high-performance modeling [274]. Furthermore, universal design frameworks utilizing refined local atomic environment descriptors, such as weighted Atomic Center Symmetry Functions (wACSF), have proven effective in predicting adsorption free energies across diverse material categories, including metal alloys, oxides, and single-atom catalysts, within a unified model [8].

A critical challenge in representation learning involves the reliance on precise atomic coordinates, which are often unavailable or computationally expensive to generate for relaxed structures. While traditional GNNs require explicit 3D coordinates, recent studies suggest that modified models can predict relaxed energies with decent precision by leveraging attention mechanisms to propagate non-geometric relative information, indicating that future research may successfully reduce reliance on exact reactant configurations [259]. To mitigate the need for exhaustive relaxation, generative models like DBCata integrate periodic Brownian-bridge frameworks with equivariant GNNs to directly generate equilibrium adsorption structures from unrelaxed inputs, achieving superior fidelity compared to traditional ML interatomic potentials [92]. The frontier of representation learning now extends into multimodal frameworks that couple graph-based geometric data with language models. Approaches such as CatBERTa and QE-Catalytic leverage large language models (LLMs) to process human-readable textual descriptions of catalyst systems, aligning latent spaces with graph neural networks to predict adsorption energies without strict dependence on 3D coordinates [Catalyst Property Prediction with CatBERTa; 161]. These multimodal models not only function as high-performance predictors but also enable inverse design by autoregressively generating structure files conditioned on target properties [170].

Despite the power of these black-box models, there is a concerted effort to infuse physical theory directly into the architecture to ensure interpretability and generalizability. The TinNet framework, for instance, integrates deep learning with d-band theory, using neural networks to output physical parameters like coupling integrals that feed into a theory module, thereby maintaining interpretability while achieving accuracy comparable to purely data-driven methods [195]. Similarly, symbolic regression techniques combined with neural networks have successfully distilled interpretable descriptors for perovskite catalysts, revealing mathematical relationships consistent with established physical principles [42; 90]. This drive for interpretability extends to identifying "materials genes" from clean experimental data, bridging the gap between theoretical modeling and experimental observations even in data-scarce regimes [95]. For specific material classes like single-atom catalysts, interpretable ML methods have identified key determinants such as coordination number and doping patterns, establishing new descriptors that integrate intrinsic properties with intermediate geometries [133]. In the realm of dual-atom catalysts, frameworks like LOCAL utilize graph-based active learning to predict stability energies directly from initial structures, navigating the extensive structural variability of these systems [82]. The convergence of these diverse strategies-standardized big data, geometric deep learning, multimodal language integration, and theory-infused architectures-establishes a robust foundation for the next generation of catalyst discovery, enabling the transition from interpolating within known chemical spaces to extrapolating across novel and complex material systems.

Advanced Representation Learning and Graph Neural Networks

The paradigm shift from hand-crafted bulk descriptors to learned atomic representations has fundamentally altered the trajectory of computational catalysis, moving the field away from reliance on global properties like the d-band center which often fail to capture site-specific nuances. Graph-based neural networks (GNNs) and message-passing architectures have emerged as the dominant framework, capable of natively operating on three-dimensional atomic coordinates under periodic boundary conditions to learn site-dependent properties directly from structure. Foundational models such as SchNet, DimeNet++, and CGCNN have demonstrated remarkable proficiency in handling permutation invariance and rotational equivariance, achieving near-DFT accuracy for adsorption energies across diverse systems ranging from quinary perovskites to high-entropy alloys [77; 274]. These architectures do not merely interpolate within known chemical spaces; they learn the underlying physics of interatomic interactions, allowing for the precise modeling of adsorbate-surface systems with a fidelity that traditional descriptor-based methods cannot match. Recent innovations have further refined these capabilities; for instance, PhAST introduces task-specific improvements in graph creation and prediction heads, significantly enhancing computational efficiency and enabling CPU-based training for scalable catalyst design without sacrificing the accuracy rivaling established architectures [52].

Despite the success of pure geometric approaches, a critical challenge persists regarding the reliance on precise atomic coordinates, which are often unavailable during the early stages of inverse design or high-throughput screening. This limitation has sparked a significant debate and subsequent innovation in multimodal learning. While some studies suggest that modified GNNs can predict relaxed energies with decent accuracy even when binding site information is removed, implying a degree of robustness to geometric omission, others argue that ignoring the relative position of the adsorbate fundamentally impairs the model's ability to distinguish between different configurations [259]. To resolve this, researchers have developed frameworks that deeply couple large language models with E(3)-equivariant graph Transformers. Models like QE-Catalytic and CatalyticMLLM inject 3D geometric information into the language channel via graph-text alignment, allowing them to function as high-performance predictors even when precise coordinates are unavailable while enabling the autoregressive generation of crystal structures for target-energy-driven design [161; 170]. Similarly, CatBERTa utilizes textual inputs to predict adsorption energies, focusing on tokens related to adsorbates and bulk composition; it achieves accuracy comparable to vanilla GNNs while offering the unique advantage of canceling systematic errors in chemically similar systems through energy difference predictions [116]. This convergence of text and graph modalities represents a strategic evolution, bridging the gap between human-readable chemical intuition and rigorous geometric deep learning [232].

Beyond static property prediction, advanced representation learning has become pivotal in automating the generation of adsorbate-surface configurations, a notorious bottleneck in high-throughput screening. The inability to efficiently sample the potential energy surface has historically limited the scope of computational studies. Frameworks like Catlas utilize pretrained graph-based models to automate the generation of these configurations, significantly improving computational efficiency and enabling the rapid screening of thousands of intermetallic candidates for reactions such as direct syngas conversion [19]. Taking this further, DBCata employs a deep generative model integrating a periodic Brownian-bridge framework with an equivariant GNN to establish low-dimensional transition manifolds between unrelaxed and DFT-relaxed structures. This approach generates high-fidelity adsorption geometries without requiring explicit energy or force information, achieving interatomic distance errors nearly three times superior to state-of-the-art machine learning potential models [92]. Complementary to this, AdsorbDiff uses conditional denoising diffusion to predict optimal adsorbate sites and orientations, offering substantial improvements in accuracy and speed over heuristic placement methods, thereby transforming the configuration search from a brute-force exercise into a guided generative process [261].

The drive for physical consistency and interpretability has also spurred the development of theory-infused models that transcend black-box predictions. TinNet integrates deep learning with the well-established d-band theory of chemisorption, ensuring that predictions are not only accurate but also physically interpretable, thereby bridging the gap between data-driven ML and fundamental chemical understanding [195]. This integration is crucial for navigating vast compositional spaces, such as those found in dual-atomic catalysts and high-entropy oxides, where active learning frameworks like LOCAL leverage graph convolutional networks and uncertainty estimates to minimize the number of required first-principles calculations [82]. Furthermore, the expansion of training data to include specialized datasets like OC22 has been instrumental, addressing the need for representations that capture intricate long-range electrostatic and magnetic interactions inherent in oxide electrocatalysts [21]. The convergence of these methodologies-multimodal alignment, generative configuration sampling, and theory infusion-signifies a maturing field where representation learning serves as a comprehensive engine for accelerated catalyst discovery, moving beyond mere prediction to enable the autonomous design of complex catalytic materials.

Integrating Text Mining and Unstructured Data

Beyond the traditional reliance on structured numerical data derived from high-throughput experiments or density functional theory (DFT) calculations, the scientific literature contains vast reservoirs of latent knowledge that can be systematically extracted to guide catalyst discovery. This capability is particularly critical in data-scarce regions where experimental labels are sparse, inconsistent, or entirely absent, such as in the exploration of compositionally complex solid solutions like high-entropy alloys and multicomponent oxides. Natural language processing (NLP) techniques have emerged as powerful tools to encode chemical compositions into latent spaces based on their co-occurrence with property concepts, effectively transforming unstructured text into actionable descriptors. A foundational approach involves representing compositions as concentration-weighted linear combinations of element embeddings derived from scientific corpora. For instance, [6] demonstrates that even lightweight distributional models like Word2Vec, trained on focused electrocatalysis abstracts, can achieve significant reductions in candidate pools for high-entropy alloys and multicomponent oxides. By projecting these embeddings onto concept directions such as "conductivity" and "dielectric," researchers can apply Pareto-front selection to filter candidates without requiring explicit electrochemical training data, often matching the performance of supervised learning methods while bypassing the need for labeled datasets. This text-mining paradigm is further supported by frameworks like MatNexus, which automate the collection and processing of scientific articles to generate vector representations suitable for machine learning, thereby streamlining the extraction of insights from the ever-growing body of materials science literature [48].

The evolution of these text-derived representations has progressed from static word embeddings to sophisticated transformer-based architectures capable of capturing complex contextual relationships. While early methods assumed linearity and independence across elements, newer approaches leverage domain-specific transformers like MatSciBERT and general-purpose large language models (LLMs) such as Qwen to encode full composition prompts. [6] compares element-wise mixing against full-composition prompts, noting that while the latter allows models to capture higher-order stoichiometric interactions directly, the simpler linear combination of element embeddings often provides a more robust baseline for aggressive candidate reduction. Contrasting with purely text-based filtering, other methodologies integrate textual reasoning with retrieval-augmented generation to navigate chemical space more dynamically. [76] illustrates how LLMs, when grounded in databases of known materials, can generate hundreds of catalyst candidates that satisfy multi-objective constraints including cost, stability, and conductivity. This approach highlights a divergence in strategy: while embedding-based filtering relies on similarity to abstract concept vectors to prune search spaces, generative and retrieval-based systems actively propose novel compositions by synthesizing domain knowledge with physical constraints. Furthermore, specialized models like CatBERTa and CataLM have been developed to predict specific catalytic properties, such as adsorption energy, directly from textual inputs, offering a bridge between human-interpretable descriptions and quantitative property prediction without the strict requirement for atomic coordinates needed by graph neural networks [Catalyst Property Prediction with CatBERTa; 16]. These models suggest that textual representations can serve as effective proxies for structural data, particularly when geometric information is incomplete or when rapid screening of vast compositional libraries is required.

Despite the promise of text-derived descriptors, challenges remain regarding the generalizability and physical fidelity of these models compared to physics-informed approaches. While [6] shows that text-based filtering can retain near-top performers in combinatorial libraries, other studies emphasize the necessity of integrating physical theories to ensure robust extrapolation. For example, [195] argues that purely data-driven models, including those based on text, may struggle with out-of-distribution generalization unless constrained by established physical laws such as d-band theory. Similarly, [213] points out that models trained on specific chemistries often fail to generalize across the broader chemical space, suggesting that text-derived embeddings must be carefully validated against diverse material systems to avoid biases inherent in the training corpus. To address these limitations, recent work has moved toward multimodal frameworks that align textual latent spaces with graph neural networks, as seen in efforts to improve adsorption energy prediction by incorporating geometric constraints into language models [232; 161]. Additionally, autonomous agents like MAESTRO and Catalyst-Agent are beginning to combine LLM reasoning with computational feedback loops, allowing for iterative hypothesis generation and validation that mitigates the risk of hallucinated chemical facts [7; 94]. Nevertheless, the consensus across recent literature indicates that integrating unstructured text data provides a unique advantage in the early stages of discovery, allowing researchers to efficiently navigate the "combinatorial explosion" of possible compositions before committing resources to expensive simulations or experiments. By leveraging the collective knowledge embedded in decades of scientific publication, text mining transforms the literature from a passive archive into an active design tool, enabling the identification of promising catalyst candidates in regions where traditional data-driven approaches falter due to label scarcity.

Challenges in Data Quality and Uncertainty Quantification

The reliability of machine learning (ML) models in catalysis is fundamentally constrained by the fidelity of input data and the rigorous quantification of prediction uncertainties, particularly when models are tasked with extrapolating to unseen chemical spaces. A primary source of discrepancy lies in the "reality gap" between computational predictions, often derived from Density Functional Theory (DFT) at 0 K, and experimental reality, which involves finite temperatures, dynamic surface restructuring, and complex reaction environments. While large-scale initiatives have attempted to bridge this divide, the divergence between simulation and experiment necessitates multi-fidelity screening protocols and robust uncertainty quantification (UQ) to build trust in AI-driven discoveries. The Open Catalyst 2020 (OC20) Dataset and Community Challenges and its successor, The Open Catalyst 2022 (OC22) Dataset and Challenges for Oxide Electrocatalysts, represent monumental efforts to provide diverse, large-scale training data comprising millions of DFT relaxations. However, even with these resources, models trained on OC20 struggle to generalize across elemental compositions and adsorbate configurations, a limitation that is particularly acute for oxide surfaces where long-range electrostatic and magnetic interactions are critical and often underrepresented in standard graph neural network architectures [21]. Furthermore, while models like PhAST have improved computational efficiency and accuracy for specific tasks through task-specific innovations in graph creation and energy prediction heads, they highlight that current architectures remain insufficient for practical, scalable applications without further innovation in handling these generalization gaps [52]. The noise inherent in experimental datasets and the scarcity of high-fidelity labeled data exacerbate these issues, necessitating UQ protocols that clearly distinguish between aleatoric uncertainty, arising from data noise, and epistemic uncertainty, reflecting model ignorance in unexplored regions [213].

Traditional black-box models often lack calibrated uncertainty estimates, leading to overconfidence in regions of chemical space devoid of training data. To address this, researchers are increasingly turning to ensemble methods and symbolic regression to construct more interpretable and reliable workflows. For instance, the SISSO-guided active learning workflow constructs ensembles of symbolic regression models to quantify mean predictions and their uncertainty, effectively identifying acid-stable oxides while mitigating the overconfidence issues common in standard bagging approaches [132]. Similarly, probabilistic approaches like Convex hull-aware Active Learning (CAL) explicitly minimize uncertainty in the convex hull of phase diagrams, prioritizing compositions that are critical for thermodynamic stability rather than merely reducing global energy errors [286]. These methods underscore a shift from purely predictive modeling to decision-making frameworks where uncertainty drives the acquisition of new data, ensuring that computational resources are allocated to the most informative regions of the design space [136].

Bridging the simulation-to-experiment gap requires transfer learning strategies that can leverage abundant low-fidelity computational data to inform sparse high-fidelity experimental observations. Transfer learning from first-principles calculations to experiments with chemistry-informed domain transformation proposes a novel Sim2Real framework that maps computational data into the experimental domain using underlying physical laws, achieving high accuracy with fewer than ten experimental data points [255]. This is crucial because experimental datasets are often biased towards known materials and suffer from inconsistencies in measurement protocols. Initiatives like the OCx24 aim to provide large, diverse experimental datasets synthesized under industrially relevant conditions to standardize data practices, while roadmap documents emphasize that reliable quantification of uncertainties remains the biggest challenge for data-centric materials modeling [236]. Without such standardized, high-quality experimental benchmarks, models risk learning artifacts of specific computational functionals or experimental setups rather than universal physical principles.

Moreover, the integration of physics into deep learning architectures offers a pathway to improve generalizability and interpretability, thereby reducing epistemic uncertainty. Theory-infused neural networks (TinNet) integrate the d-band theory of chemisorption directly into the network architecture, ensuring that predictions adhere to established physical laws even when extrapolating to unseen systems [195]. Similarly, models like DOTA leverage local density of states to predict adsorption energies with chemical accuracy, resolving long-standing challenges such as the "CO puzzle" that purely data-driven models often miss [138]. Despite these advances, contradictions remain regarding the necessity of explicit geometric information; while some studies suggest that removing binding site information significantly impairs accuracy, others demonstrate that modified models can predict relaxed energies with decent performance by propagating non-geometric relative information, suggesting potential avenues for reducing data requirements [259].

Multimodal approaches like QE-Catalytic further attempt to resolve this by coupling language models with geometric graph transformers, yet the tension between data efficiency and physical rigor persists [161]. Ultimately, bridging the gap between computational predictions and experimental reality requires a synergistic approach combining large-scale, diverse datasets, rigorous uncertainty quantification, physics-informed architectures, and autonomous workflows that iteratively refine models based on experimental feedback [40; 39].

Supervised Learning, Predictive Modeling, and Kinetic Analysis

Supervised learning has firmly established itself as a cornerstone methodology in the rational design of electrocatalysts, fundamentally altering the landscape of how thermodynamic and kinetic parameters are predicted. Traditionally, obtaining these parameters required computationally expensive density functional theory (DFT) calculations, which limited the scope of high-throughput screening. Modern ensemble models have effectively bridged this gap by mapping complex compositional and structural features to catalytic performance metrics with remarkable fidelity. For instance, stacking ensemble models that integrate Random Forest, XGBoost, and Support Vector Regression have demonstrated exceptional efficacy in predicting the Gibbs free energy of adsorption for both the Hydrogen Evolution Reaction (HER) and Oxygen Evolution Reaction (OER). By leveraging sophisticated feature engineering strategies, including Matminer-based compositional analysis and Principal Component Analysis, these models have achieved R^2 values of 0.98 and 0.94 respectively on large-scale datasets, validating the power of combining diverse algorithms to capture non-linear structure-property relationships [2]. Similarly, in the realm of complex solid solutions, convolutional neural networks (CNNs) have been employed to navigate the near-infinite design space of high-entropy alloys. These deep learning architectures successfully capture non-linear correlations in adsorption energies for ammonia synthesis intermediates, achieving mean absolute errors as low as 0.05 eV, thereby proving that deep learning can effectively model the intricate atomic environments found in disordered materials [120].

While black-box models offer superior predictive power, a significant paradigm shift is occurring toward interpretability and the derivation of explicit mathematical relationships between descriptors and activity. The community is increasingly recognizing that accurate prediction alone is insufficient for scientific discovery; understanding the underlying physics is paramount. Symbolic regression techniques, such as the Sure-Independence Screening and Sparsifying Operator (SISSO), are now being utilized to bridge the gap between data-driven predictions and physical understanding. By generating analytical expressions, these methods identify key physical parameters governing properties like acid stability in oxides or adsorption energies, often revealing relationships that are consistent with established physics-based equations, such as the direct proportionality between the coordination number of the adsorbing atom and adsorption energy [90; 132]. This approach not only enhances model transparency but also facilitates active learning workflows where uncertainty estimates guide the selection of new materials for high-fidelity calculation, significantly reducing the computational cost of discovery. Furthermore, theory-infused neural networks represent a hybrid paradigm where deep learning architectures are constrained by physical laws. Models like TinNet incorporate scientific knowledge, such as the d-band theory of chemisorption, directly into the network structure, ensuring that predictions remain physically interpretable while maintaining the accuracy of purely data-driven approaches [195].

The scope of machine learning in catalysis has expanded beyond static property prediction to encompass dynamic microkinetic modeling and the prediction of reaction rate constants. This evolution allows for a more nuanced assessment of catalytic performance that accounts for surface coverage effects and competing reaction pathways, addressing the limitations of static descriptors like the Gibbs free energy of adsorption. Machine learning models have been successfully trained to predict mass activity for oxygen reduction reactions by leveraging microkinetic models that estimate site contributions based on generalized coordination number distributions [162]. In this context, Bayesian optimization coupled with Gaussian Process Regression has proven effective in identifying top-performing nanoparticle structures from pools exceeding 50,000 candidates, demonstrating the capability of these frameworks to navigate complex structure-activity landscapes efficiently. Moreover, recent advancements include the use of machine learning to predict quantum flux-flux correlation functions and reaction rate constants directly, offering significant speedups over traditional time-propagation methods while maintaining high accuracy for organic heterogeneous catalytic surface reactions [128].

Despite these successes, challenges remain regarding the generalizability of models across diverse chemical spaces and the integration of dynamic surface phenomena. Large-scale datasets like the Open Catalyst 2020 (OC20) and OC22 have been curated to address the scarcity of training data, particularly for oxide electrocatalysts, pushing the community toward building universal machine learning potentials [77; 21]. However, models often struggle to generalize to underperforming subsets or chemically dissimilar materials without specialized fine-tuning [213]. Additionally, the dynamic nature of electrocatalyst surfaces under operando conditions necessitates models that can account for potential-driven restructuring and coverage-dependent activity. Recent work utilizing coverage-aware machine learning potentials has begun to resolve these operando interfaces, revealing that catalytic activity is governed by statistical ensembles of interconverting adsorbate motifs rather than single average-coverage sites [131]. This highlights the critical need for future models to integrate microkinetic insights with dynamic structural representations to fully capture the complexity of real-world catalytic environments.

High-Throughput Screening and Ensemble Methods

The paradigm shift in electrocatalyst discovery from intuition-driven trial-and-error to data-driven high-throughput screening (HTS) has been catalyzed by the synergistic integration of machine learning (ML) predictions with rigorous density functional theory (DFT) verification. This coupled workflow addresses the combinatorial explosion inherent in exploring vast compositional spaces, such as those found in bimetallic alloys, high-entropy materials, and complex nanostructures, which were previously inaccessible to purely computational or experimental approaches. A cornerstone of this strategy is the deployment of stacking ensemble models, which aggregate the predictive strengths of diverse algorithms to achieve robustness and accuracy superior to single-model architectures. For instance, a comprehensive study utilizing a dataset of over 16,000 points from the Catalysis-Hub database demonstrated that a stacking ensemble combining Random Forest, XGBoost, and Support Vector Regression could predict Gibbs free energy of adsorption with exceptional fidelity, achieving R^2 values of 0.98 for the Hydrogen Evolution Reaction (HER) and 0.94 for the Oxygen Evolution Reaction (OER) [2]. By leveraging advanced feature engineering strategies, including Matminer-based compositional analysis and Principal Component Analysis, these models generate robust descriptors that effectively bridge the gap between abstract data patterns and fundamental electrochemical principles, enabling the pre-screening of candidates before committing to costly DFT relaxations.

Beyond static predictive modeling, the evolution of HTS workflows has increasingly incorporated dynamic search strategies like Bayesian optimization and genetic algorithms to navigate complex, multi-dimensional design spaces efficiently. In the realm of compositionally complex solid solutions, such as high-entropy alloys, the sheer number of possible compositions renders grid-search methods impractical. Bayesian optimization has proven instrumental in this context, successfully identifying global activity optima in quaternary systems like Ni-Pd-Pt-Ru for OER after sampling less than 20% of the complete composition space [83]. This approach treats catalyst discovery as an iterative optimization problem, balancing exploration of unknown regions with exploitation of known high-performing areas. Similarly, Bayesian genetic algorithms have been employed to simultaneously optimize material composition and experimental conditions, a capability critical for translating theoretical predictions into practical devices. This methodology led to the identification of Ni-doped carbon quantum dots (Ni@CQD) as a superior HER catalyst, with experimental validation confirming an overpotential of 189 mV at 10 mA cm^{-2}, thereby validating the utility of data-driven screening in identifying earth-abundant alternatives to precious metals [10].

The efficacy of these workflows is further enhanced by multi-stage discovery processes that integrate active learning and symbolic regression to refine search spaces iteratively, particularly in data-scarce regimes where deep learning models may struggle. A notable example involves the discovery of advanced acidic OER catalysts, where a multi-stage approach combining data mining, active learning, and domain adaptation systematically narrowed the exploration space to identify a promising Ru-Mn-Ca-Pr oxide catalyst [3]. This contrasts sharply with traditional methods reliant on subjective intuition, marking a shift toward systematic, data-informed exploration. Furthermore, symbolic regression techniques, such as the Sure-Independence Screening and Sparsifying Operator (SISSO), have been utilized within active learning frameworks to identify acid-stable oxides. By constructing ensembles of SISSO models to quantify uncertainty, researchers successfully identified 12 acid-stable materials from a pool of 1,470 oxides in only 30 iterations, showcasing the efficiency of interpretable models in guiding high-fidelity DFT calculations [132].

The scalability of these HTS methodologies extends well beyond metallic alloys to encompass complex nanostructures, single-atom catalysts (SACs), and two-dimensional materials like MXenes. Machine learning potentials (MLPs) have emerged as a critical tool in this expansion, offering near-DFT accuracy at a fraction of the computational cost, thus enabling the screening of thousands of coordination motifs that would be prohibitive with standard DFT. For example, MLPs were utilized to map the HER landscape of graphene-supported dual-atom catalysts, identifying standout candidates like Ti2@2Na and Mn2@2Na from a vast design space [5]. In the context of MXenes, a multistep workflow combining DFT and supervised ML algorithms screened over 4,500 configurations to discover highly stable and active catalysts surpassing commercial platinum counterparts [50]. The magnitude of these efforts is exemplified by the screening of perovskites, where ML models based on elemental features predicted catalytic properties for over 19 million candidates, identifying mixtures of less-explored elements as top performers [74].

Underpinning these diverse applications is the availability of large-scale, standardized datasets that improve model generalizability across different adsorbates and surface compositions. Initiatives like the Open Catalyst 2020 (OC20) and Open Catalyst 2022 (OC22) datasets have been pivotal in addressing data scarcity, providing millions of DFT relaxations that serve as the foundation for training robust graph neural networks and other advanced architectures [77] [21]. Recent advancements also include the use of generative models to predict relaxed adsorption structures directly, bypassing expensive relaxation steps and further accelerating the screening pipeline [92]. The convergence of stacking ensembles, Bayesian optimization, active learning, and generative modeling establishes a robust framework for the rational design of next-generation electrocatalysts. This framework effectively balances the trade-off between computational cost and predictive accuracy, ensuring that findings are not only theoretically sound but also validated through experimental synthesis and characterization, thereby closing the loop between simulation and reality.

Machine Learning Potentials and Force Fields

The prohibitive computational expense of ab initio molecular dynamics and large-scale structural relaxations has historically constrained the exploration of vast chemical spaces in heterogeneous catalysis, limiting studies to small unit cells and static configurations. To overcome these barriers, machine learning potentials (MLPs) and machine-learned force fields (MLFFs) have emerged as transformative tools capable of approximating Density Functional Theory (DFT) accuracy at a fraction of the computational cost. These models facilitate the screening of thousands of coordination motifs and surface configurations, enabling researchers to probe complex systems that were previously intractable. For instance, in the design of nitrogen-coordinated dual-atom catalysts for the hydrogen evolution reaction, MLPs have been successfully deployed to screen 23 transition metals across 20 nitrogen coordination motifs, achieving mean absolute errors as low as 80 meV for Gibbs binding free energies while identifying numerous synthesizable candidates [5]. Similarly, MLFFs have proven essential in modeling dynamic surface evolution under reactive conditions, such as nanoparticle restructuring and CO2-to-methanol conversion, where they enable the proposal of novel descriptors like adsorption energy distribution [13]. This descriptor approach aggregates binding energies across diverse facets and sites, offering a more holistic view of nanoparticle catalysis than traditional single-site models [43].

A critical advancement in this domain is the development of universal datasets and models that generalize across diverse elemental compositions and adsorbate identities, moving beyond system-specific potentials. The Open Catalyst 2020 (OC20) dataset, comprising over 1.2 million DFT relaxations, served as a foundational benchmark for training graph neural networks to predict energies and forces across a wide swath of materials [77]. Building upon this, recent efforts have extended these capabilities to solid-liquid interfaces through the Open Catalyst 2025 (OC25) dataset, which includes over 7.8 million calculations spanning explicit solvent environments and electrolytes. This expansion addresses the limitations of gas-phase models by capturing the intricate long-range electrostatic and magnetic interactions inherent in electrocatalytic systems, allowing state-of-the-art models to achieve energy errors as low as 0.1 eV [68]. Furthermore, specialized datasets for oxide electrocatalysts, such as OC22, have been curated to improve model performance on materials critical for the oxygen evolution reaction, demonstrating that joint training across chemically dissimilar datasets can significantly enhance prediction accuracy [21]. These datasets provide the necessary diversity to train robust MLFFs that can handle the complexity of real-world catalytic environments, including the dynamic interplay between solvents, ions, and surfaces.

The efficacy of MLFFs extends beyond static property prediction to dynamic simulations and global structure optimization, challenging conventional wisdom regarding model fidelity. Surprisingly, studies indicate that learned force fields can locate ground state structures with energies comparable to or lower than those found by traditional DFT relaxations, even when the predicted forces exhibit significant errors relative to the ground truth. This phenomenon suggests that the primary requirement for structure optimization is the correct identification of energy minima rather than perfect force fidelity, a finding that accelerates the discovery of stable catalyst configurations [281]. In fact, force fields trained on simple harmonic approximations around minima, termed "Easy Potentials," can converge faster and find lower energy structures than models trained on full DFT forces, illustrating that the topology of the potential energy surface near the minimum is more critical than global force accuracy. In the context of high-entropy alloys and complex nanoalloys, MLFFs trained on modified embedded-atom method data have enabled rapid homotop optimization via Monte Carlo simulations, facilitating the mapping of catalytically reactive hotspots based on d-band centers without the prohibitive cost of full electronic structure calculations for every configuration [276]. Additionally, these potentials have been instrumental in simulating high-temperature processes, such as the pyrolysis of metal-organic frameworks to form amorphous catalysts, providing atomistic insights into gas evolution and metal nanoparticle formation under experimentally relevant conditions [143].

The integration of MLFFs with generative and agentic workflows represents a paradigm shift in how adsorption configurations are identified and optimized. Traditional methods often rely on exhaustive sampling or heuristic placement, which may miss the global minimum energy configuration. To address this, generative models like AdsorbDiff utilize conditional denoising diffusion to predict optimal adsorbate sites and orientations, achieving significant improvements in accuracy and speed compared to brute-force approaches [261]. Furthermore, autonomous agents leveraging large language models, such as Adsorb-Agent, strategically explore configuration spaces by combining reasoning capabilities with fast MLFF evaluations, significantly reducing the number of required initial setups while improving the reliability of energy predictions [244]. These advancements collectively underscore the transition from purely data-driven interpolation to physics-informed, generative, and autonomous frameworks that are reshaping the landscape of computational catalyst discovery. By enabling the rapid evaluation of complex, dynamic, and disordered systems, MLFFs are not merely accelerating existing workflows but are unlocking entirely new avenues for exploring the catalytic landscape, from breaking scaling relations in inverse catalysts to designing high-entropy materials with tailored active sites. The synergy between accurate force prediction, robust dataset curation, and intelligent sampling strategies now allows for the simulation of catalytic phenomena at time and length scales that were previously the exclusive domain of classical force fields, but with the quantum mechanical accuracy required for reliable mechanistic insight.

Limitations of Thermodynamic Descriptors and Kinetic Refinements

The reliance on thermodynamic descriptors, particularly adsorption energy, as the primary screening criterion for catalyst discovery has come under increasing scrutiny due to inherent uncertainties in Density Functional Theory (DFT) predictions and the frequent neglect of kinetic barriers. While adsorption energy serves as a foundational reactivity descriptor, determining the global minimum energy configuration requires evaluating numerous adsorbate-catalyst geometries, a process that is computationally intensive and prone to error if exhaustive sampling is not performed [244]. Even with advanced sampling, the uncertainty in adsorption free energy calculations can lead to the misclassification of materials, suggesting that static thermodynamic models are insufficient for capturing the full complexity of catalytic performance. Recent advancements in generative models, such as those integrating periodic Brownian-bridge frameworks with equivariant graph neural networks, have demonstrated that directly generating equilibrium adsorption structures can significantly reduce errors in adsorption energy predictions, yet the fundamental limitation remains: a thermodynamically favorable adsorption energy does not guarantee catalytic activity if kinetic barriers are prohibitive [92].

This limitation is starkly evident in the Hydrogen Evolution Reaction (HER), where the seminal N?rskov model, which posits a universal rate constant and a volcano relationship based solely on hydrogen adsorption free energy (ΔG_H), often deviates from experimental exchange current densities by up to six orders of magnitude [183]. Recent analyses indicate that this discrepancy arises because the kinetic pre-factor is not universal but rather depends linearly on the absolute value of ΔG_H, necessitating a refinement of classical kinetic models to include material-specific activation entropies and outer-sphere effects [183]. The assumption that a single thermodynamic descriptor can universally predict activity across diverse material classes is further challenged by emerging frameworks that incorporate dynamic and kinetic factors. For instance, the anomalous pH dependence observed in hydrogen electrocatalysis cannot be explained by traditional Nernstian models or static adsorption energies alone. Instead, a waveguide kinetics framework has been introduced to reinterpret polarization curves as power-flow responses, offering a unified description of interfacial efficiency that accounts for mechanistic nuances missed by thermodynamic volcanoes [272].

Similarly, in complex multi-step reactions like dry reforming of methane or the oxygen evolution reaction (OER), the rate-determining step is often governed by kinetic bottlenecks such as intermediate saturation or specific activation barriers rather than the thermodynamic stability of the final products. Microkinetic modeling of Pt-catalyzed dry reforming reveals that CO desorption and intermediate regeneration steps create kinetic traps that static descriptors fail to identify, highlighting the necessity of automated chemical kinetic model generation to capture operational regimes [218]. In the context of OER, estimating reaction rate constants directly from experimental impedance spectra provides a more accurate identification of the rate-determining step than relying solely on DFT-computed Gibbs free energies, which often assume idealized configurations at zero Kelvin [148]. The integration of kinetic analysis also extends to the stability of catalysts under operating conditions, where thermodynamic descriptors alone may predict active materials that are unstable in realistic environments. The use of Pourbaix diagrams and constant potential reactor frameworks allows for the simulation of electrified solid-liquid interfaces over extended timescales, uncovering mechanisms such as cation-promoted CO_2 adsorption that are invisible to static vacuum calculations [200].

Moreover, the stability of high-entropy alloys and complex surface structures often depends on dynamic phenomena like surface segregation, which is driven by thermodynamic factors such as surface energy differences but manifests through kinetic evolution at operational temperatures. Molecular dynamics simulations combined with machine learning force fields have demonstrated that surface segregation can significantly alter the availability of active sites, rendering predictions based on ideal bulk terminations inaccurate [249]. To address these challenges, recent advancements employ machine learning interatomic potentials and active learning strategies to accelerate the exploration of potential energy surfaces, ensuring that both thermodynamic stability and kinetic accessibility are considered simultaneously [167]. The move toward kinetic refinements is further supported by the development of interpretable machine learning models that infuse physical theories, such as d-band theory, with data-driven approaches to predict reactivity more accurately than pure thermodynamic descriptors. These theory-infused neural networks can capture non-linear correlations and electronic structure effects that simple scaling relations miss, providing a more robust framework for identifying rate-determining steps and optimal catalytic motifs [195]. Additionally, the application of Br?nsted-Evans-Polanyi (BEP) relations in conjunction with machine learning has allowed for the prediction of activation energies across vast chemical spaces, bridging the gap between thermodynamic driving forces and kinetic barriers [13]. Ultimately, the field is shifting from a paradigm dominated by static adsorption energies to one that embraces unified kinetic models, microkinetic simulations, and dynamic stability assessments to achieve truly predictive catalyst design.

Generative Models, Inverse Design, and Autonomous Agents

The paradigm of catalyst discovery is undergoing a fundamental transformation, shifting from passive data analysis and predictive screening to active hypothesis generation and autonomous design. Traditional approaches, often reliant on trial-and-error or static high-throughput screening of predefined libraries, are increasingly being supplanted by generative models and autonomous agents capable of navigating vast chemical spaces to propose novel structures and optimize synthesis parameters. This evolution is exemplified by the emergence of Large Language Model (LLM)-powered agents that operationalize complex workflows. For instance, Catalyst-Agent demonstrates the capability of an LLM-based agent to explore material databases, perform structural modifications, and calculate adsorption energies in a closed-loop manner, achieving significant success rates in identifying candidates for oxygen, nitrogen, and CO2 reduction reactions with minimal human intervention [94]. Similarly, Adsorb-Agent leverages the reasoning capabilities of LLMs to strategically identify stable adsorption configurations, significantly reducing the computational cost associated with exhaustive sampling while improving prediction accuracy [244]. These systems represent a move toward "self-driving" laboratories where agentic AI not only executes tasks but also derives interpretable design rules, as seen in ChemNavigator, which autonomously identified statistically significant structure-property relationships for organic photocatalysts through iterative hypothesis testing [56].

Beyond agent-based orchestration, generative models are enabling true inverse design, where materials are constructed de novo to meet specific target properties rather than selected from existing datasets. Generative Flow Networks (GFlowNets) and diffusion models have shown particular promise in exploring complex energy landscapes. Catalyst GFlowNet utilizes machine learning predictors to design crystal surfaces for the hydrogen evolution reaction, successfully identifying known optimal catalysts like platinum while opening the search space for new materials [1]. Diffusion-based approaches are also advancing; DBCata integrates a periodic Brownian-bridge framework with equivariant graph neural networks to directly generate equilibrium adsorption structures, offering superior accuracy over traditional machine learning interatomic potentials [92]. Furthermore, DOSMatGen introduces an E(3)-equivariant joint diffusion framework capable of generating crystal structures that match a desired electronic density of states, allowing for physics-driven inverse design targeting specific functional properties [263]. To ensure chemical validity during generation, frameworks like CrysVCD integrate valence constraints directly into the generative process, significantly improving the thermodynamic stability of generated candidates compared to post-screening methods [212].

A critical advantage of these advanced frameworks is their ability to break conventional scaling relations that limit traditional catalyst design. Linear scaling relations between adsorption energies of different intermediates often impose a theoretical ceiling on catalytic activity. However, multi-agent LLM frameworks like MAESTRO utilize iterative reasoning and in-context learning to discover single-atom catalysts that defy these scaling relations, identifying design principles not explicitly encoded in their training data [7]. This capability is further supported by hierarchical generative modeling, which couples genetic algorithms for configurational search with generative models for molecular design. This approach allows for the simultaneous refinement of geometry and composition in multi-component systems, successfully reducing activation barriers in catalytic environments by optimizing surrounding components around a fixed transition state [58]. Additionally, inverse design frameworks such as MAGECS combine bird swarm algorithms with graph neural networks to navigate the global chemical space, generating hundreds of thousands of promising alloy structures for CO2 reduction that exhibit high activity and have been experimentally validated [Inverse Design of Promising Alloys for Electrocatalytic $\mathbf { C O } _ { 2 }$ Reduction via Generative Graph Neural Networks Combined with Bird Swarm Algorithm].

The integration of reasoning-driven search with quantum-chemical feedback further enhances the reliability of these discoveries. CHEMREASONER formulates catalyst discovery as an uncertain environment where an agent actively searches for effective catalysts by combining LLM-derived hypotheses with atomistic graph neural network feedback, steering exploration toward energetically favorable regions without human input [CHEMREASONER: Heuristic Search over a Large Language Model's Knowledge Space using Quantum-Chemical Feedback]. This synergy between linguistic reasoning and physical constraints is also evident in retrieval-augmented generation frameworks, which enable LLMs to navigate chemical spaces grounded in databases of known materials, generating high-entropy catalyst candidates with high thermodynamic stability and optimized cost-performance trade-offs [76]. Moreover, unified multimodal models like CatalyticMLLM and QE-Catalytic integrate property prediction and inverse design within a single framework, aligning graph and text representations to reduce distribution shifts and improve the stability of closed-loop optimization workflows [170] [161]. These developments collectively signify a shift where AI does not merely accelerate screening but actively participates in the scientific method, proposing, testing, and refining hypotheses to uncover catalysts that transcend conventional design limitations.

Conclusion and Future Perspectives

The landscape of electrocatalyst discovery has undergone a fundamental transformation, evolving from a discipline reliant on serendipitous experimentation and intuition-driven heuristics to one defined by rational, data-centric design. This paradigm shift is characterized by the convergence of high-throughput computational screening, advanced machine learning potentials, and autonomous experimental workflows, which collectively mitigate the combinatorial explosion of chemical space and the prohibitive costs associated with traditional density functional theory (DFT). The establishment of foundational datasets and universal machine learning models has not only accelerated the identification of earth-abundant alternatives to noble metals but has also facilitated the exploration of complex material classes, including high-entropy alloys, single-atom catalysts, and topological quantum materials. These advancements underscore the capacity of artificial intelligence (AI) to uncover non-intuitive design principles that transcend conventional linear scaling relations, thereby pushing the theoretical limits of catalytic efficiency.

Despite these significant strides, the field faces persistent challenges that necessitate a shift from purely predictive modeling to holistic, physics-informed frameworks. The reliability of thermodynamic descriptors alone has proven insufficient for identifying superior catalysts, as static models often fail to capture the dynamic nature of active sites under operating conditions. Consequently, the community is increasingly adopting multi-fidelity workflows that integrate kinetic modeling, stability analysis, and synthesizability constraints. To bridge the critical divide between simulation and experiment, researchers are leveraging active learning strategies and self-driving laboratories that create virtuous cycles of data generation and model refinement. Furthermore, the integration of explainable AI and theory-infused neural networks is proving essential for distilling complex model outputs into physically meaningful insights, ensuring that data-driven discoveries remain grounded in fundamental chemical principles and are not merely artifacts of black-box correlations.

Looking forward, the trajectory of electrocatalysis points toward the emergence of fully autonomous discovery ecosystems powered by agentic AI. These systems promise to operationalize the scientific method by autonomously hypothesizing, planning, and executing discovery workflows, effectively transforming AI from a passive predictive tool into an active scientific collaborator. The future of sustainable energy conversion relies on the successful maturation of these integrated frameworks, which must seamlessly couple atomic-scale precision with multiscale engineering and robust experimental validation. By democratizing access to advanced design tools and accelerating the deployment of high-performance catalysts, this evolving landscape holds the potential to deliver the scalable solutions required to address global climate challenges and achieve carbon neutrality. The ultimate goal is no longer just the acceleration of discovery, but the establishment of a continuous, self-improving loop of knowledge generation that can adapt to the complex demands of next-generation energy technologies.